Hawking, with the conclusion that the maximum entropy potential for a region of space is proportional to its surface area, not its volume.

All right, but why should we care? There are two reasons.

First, the entropy bound gives yet another clue that ultramicroscopic space has an atomized structure. In detail, Bekenstein and Hawking found that if you imagine drawing a checkerboard pattern on the event horizon of a black hole, with each square being one Planck length by one Planck length (so each such "Planck square" has an area of about 10^{-66} square centimeters), then the black hole's entropy equals the number of such squares that can fit on its surface.[4] It's hard to miss the conclusion to which this result strongly hints: each Planck square is a minimal, fundamental unit of space, and each carries a minimal, single unit of entropy. This suggests that there is nothing, even in principle, that can take place *within* a Planck square, because any such activity could support disorder and hence the Planck square could contain more than the single unit of entropy found by Bekenstein and Hawking. Once again, then, from a completely different perspective we are led to the notion of an elemental spatial entity.[5]

Second, for a physicist, the upper limit to the entropy that can exist in a region of space is a critical, almost sacred quantity. To understand why, imagine you're working for a behavioral psychiatrist, and your job is to keep a detailed, moment-to-moment record of the interactions between groups of intensely hyperactive young children. Every morning you pray that the day's group will be well behaved, because the more bedlam the children create, the more difficult your job. The reason is intuitively obvious, but it's worth saying explicitly: the more disorderly the children are, the more things you have to keep track of. The universe presents a physicist with much the same challenge. A fundamental physical theory is meant to describe everything that goes on—or could go on, even in principle—in a given region of space. And, as with the children, the more disorder the region can contain—even in principle—the more things the theory must be capable of keeping track of. Thus, the maximum entropy a region can contain provides a simple but incisive litmus test: physicists expect that a truly fundamental theory is one that is perfectly matched to the maximum entropy in any given spatial region. The theory should be so tightly in tune with nature that its maximum